

SYNTH-OEM-A diagnostic specification

OEM OEM-A | version 1.0

ecu_name: "SYNTH-OEM-A"

oem: "OEM-A"

version: "1.0"

Data identifiers				
DID	Name	Bytes	Read sessions	Write sessions
0x1000	VIN	4	1, 3	3
0x1001	DID-1	4	1, 3	3
0x1002	DID-2	4	1, 3	3
0x1003	DID-3	4	1, 3	3
0x1004	DID-4	4	1, 3	3

Data identifiers				
DID	Name	Bytes	Read sessions	Write sessions
0x1005	DID-5	4	1, 3	3
0x1006	DID-6	4	1, 3	3
0x1007	DID-7	4	1, 3	3
0x1008	DID-8	4	1, 3	3
0x1009	DID-9	4	1, 3	3

Prose appendix: operating conditions

DID 0x1000 may be written only when signal_0 is true.

Writing DID 0x1000 requires signal_0 to be true.

DID 0x1000 must not be written while signal_0 is false.

The VIN data identifier may be written only when signal_0 is true.

DID 0x1001 may be written only when vehicle_speed is below 10 km/h.

Writing DID 0x1001 requires vehicle_speed to be under 10 km/h.

DID 0x1001 must not be written while vehicle_speed is not below 10 km/h.

The DID-1 data identifier requires vehicle_speed below 10 km/h.

DID 0x1002 may be written only when battery_voltage is below 14 V.

Writing DID 0x1002 requires battery_voltage to be under 14 V.

DID 0x1002 must not be written while battery_voltage is not below 14 V.

The DID-2 data identifier requires battery_voltage below 14 V.

DID 0x1003 may be written only when response_delay is below 50 ms.

Writing DID 0x1003 requires response_delay to be under 50 ms.

DID 0x1003 must not be written while response_delay is not below 50 ms.

The DID-3 data identifier requires response_delay below 50 ms.

DID 0x1004 may be written only when engine_temp is below 95 °C.

Writing DID 0x1004 requires engine_temp to be under 95 °C.

DID 0x1004 must not be written while engine_temp is not below 95 °C.

The DID-4 data identifier requires engine_temp below 95 °C.

DID 0x1005 may be written only when signal_5 is true and ignition_on is true.

DID 0x1005 may be written only when signal_5 is true.

Writing DID 0x1005 requires signal_5 to be true.

DID 0x1005 must not be written while signal_5 is false.

The DID-5 data identifier may be written only when signal_5 is true.

DID 0x1005 may be written only when ignition_on is true.

Writing DID 0x1005 requires ignition_on to be true.

DID 0x1005 must not be written while ignition_on is false.

The DID-5 data identifier may be written only when ignition_on is true.

```
sessions: [{ 'id': 1, 'name': 'default'}, { 'id': 3, 'name': 'extended'}]
```

```
sessions[0].id: 1
```

```
sessions[0].name: "default"
```

```
sessions[1].id: 3
```

```
sessions[1].name: "extended"
```

```
security_levels: [{ 'level': 0, 'name': 'locked'}, { 'level': 1, 'name': 'level1'}]
```

```
security_levels[0].level: 0
```

```
security_levels[0].name: "locked"
```

```
security_levels[1].level: 1
```

```
security_levels[1].name: "level1"
```

```
services: [{ 'sid': 16, 'name': 'SessionControl', 'allowed_sessions': [], 'required_security_level': None, 'subfunctions': [],  
'suppress_positive_response_supported': False}, { 'sid': 17, 'name': 'Reset', 'allowed_sessions': [],  
'required_security_level': None, 'subfunctions': [{ 'value': 1, 'name': 'hard', 'allowed_sessions': [],  
'required_security_level': None}], 'suppress_positive_response_supported': False}, { 'sid': 34, 'name':  
'ReadDataByIdentifier', 'allowed_sessions': [], 'required_security_level': None, 'subfunctions': [],  
'suppress_positive_response_supported': False}, { 'sid': 39, 'name': 'SecurityAccess', 'allowed_sessions': [],  
'required_security_level': None, 'subfunctions': [], 'suppress_positive_response_supported': False}, { 'sid': 46, 'name':  
'WriteDataByIdentifier', 'allowed_sessions': [], 'required_security_level': None, 'subfunctions': [],  
'suppress_positive_response_supported': False}, { 'sid': 49, 'name': 'RoutineControl', 'allowed_sessions': [3],  
'required_security_level': 1, 'subfunctions': [], 'suppress_positive_response_supported': False}, { 'sid': 62, 'name':  
'TesterPresent', 'allowed_sessions': [], 'required_security_level': None, 'subfunctions': [],  
'suppress_positive_response_supported': True}]
```

```
services[0].sid: 16
```

```
services[0].name: "SessionControl"
```

```
services[0].allowed_sessions: []
```

```
services[0].required_security_level: null
```

```
services[0].subfunctions: []
```

```
services[0].suppress_positive_response_supported: false
```

```
services[1].sid: 17
```

```
services[1].name: "Reset"
```

```
services[1].allowed_sessions: []
```

```
services[1].required_security_level: null
```

```
services[1].subfunctions: [{"allowed_sessions": [], "name": "hard", "required_security_level": null, "value": 1}]
```

```
services[1].suppress_positive_response_supported: false
```

```
services[2].sid: 34
```

```
services[2].name: "ReadDataByIdentifier"
services[2].allowed_sessions: []
services[2].required_security_level: null
services[2].subfunctions: []
services[2].suppress_positive_response_supported: false
services[3].sid: 39
services[3].name: "SecurityAccess"
services[3].allowed_sessions: []
services[3].required_security_level: null
services[3].subfunctions: []
services[3].suppress_positive_response_supported: false
services[4].sid: 46
services[4].name: "WriteDataByIdentifier"
services[4].allowed_sessions: []
services[4].required_security_level: null
services[4].subfunctions: []
services[4].suppress_positive_response_supported: false
services[5].sid: 49
services[5].name: "RoutineControl"
services[5].allowed_sessions: [3]
services[5].required_security_level: 1
services[5].subfunctions: []
services[5].suppress_positive_response_supported: false
services[6].sid: 62
services[6].name: "TesterPresent"
services[6].allowed_sessions: []
services[6].required_security_level: null
services[6].subfunctions: []
services[6].suppress_positive_response_supported: true
dids: [{'did': 4096, 'name': 'VIN', 'length_bytes': 4, 'encoding': 'bytes', 'read_sessions': [1, 3], 'write_sessions': [3],
'read_security': 1, 'write_security': 1, 'preconditions': [{'kind': 'signal', 'signal': 'signal_0', 'op': '==', 'value': True, 'unit':
None, 'target': None, 'source_text': 'signal_0 is true'}]}, {'did': 4097, 'name': 'DID-1', 'length_bytes': 4, 'encoding': 'bytes',
'read_sessions': [1, 3], 'write_sessions': [3], 'read_security': 1, 'write_security': 1, 'preconditions': [{'kind': 'signal', 'signal':
'vehicle_speed', 'op': '<', 'value': 10, 'unit': 'km/h', 'target': None, 'source_text': 'vehicle_speed is below 10 km/h'}]}, {'did':
4098, 'name': 'DID-2', 'length_bytes': 4, 'encoding': 'bytes', 'read_sessions': [1, 3], 'write_sessions': [3], 'read_security': 1,
'write_security': 1, 'preconditions': [{'kind': 'signal', 'signal': 'battery_voltage', 'op': '<', 'value': 14, 'unit': 'V', 'target': None,
'source_text': 'battery_voltage is below 14 V'}]}, {'did': 4099, 'name': 'DID-3', 'length_bytes': 4, 'encoding': 'bytes',
'read_sessions': [1, 3], 'write_sessions': [3], 'read_security': 1, 'write_security': 1, 'preconditions': [{'kind': 'signal', 'signal':
```

'response_delay', 'op': '<', 'value': 50, 'unit': 'ms', 'target': None, 'source_text': 'response_delay is below 50 ms'}], {'did': 4100, 'name': 'DID-4', 'length_bytes': 4, 'encoding': 'bytes', 'read_sessions': [1, 3], 'write_sessions': [3], 'read_security': 1, 'write_security': 1, 'preconditions': [{'kind': 'signal', 'signal': 'engine_temp', 'op': '<', 'value': 95, 'unit': '°C', 'target': None, 'source_text': 'engine_temp is below 95 °C'}]}, {'did': 4101, 'name': 'DID-5', 'length_bytes': 4, 'encoding': 'bytes', 'read_sessions': [1, 3], 'write_sessions': [3], 'read_security': 1, 'write_security': 1, 'preconditions': [{'kind': 'signal', 'signal': 'signal_5', 'op': '==', 'value': True, 'unit': None, 'target': None, 'source_text': 'signal_5 is true'}, {'kind': 'signal', 'signal': 'ignition_on', 'op': '==', 'value': True, 'unit': None, 'target': None, 'source_text': 'ignition_on is true'}]}, {'did': 4102, 'name': 'DID-6', 'length_bytes': 4, 'encoding': 'bytes', 'read_sessions': [1, 3], 'write_sessions': [3], 'read_security': 1, 'write_security': 1, 'preconditions': []}, {'did': 4103, 'name': 'DID-7', 'length_bytes': 4, 'encoding': 'bytes', 'read_sessions': [1, 3], 'write_sessions': [3], 'read_security': 1, 'write_security': 1, 'preconditions': []}, {'did': 4104, 'name': 'DID-8', 'length_bytes': 4, 'encoding': 'bytes', 'read_sessions': [1, 3], 'write_sessions': [3], 'read_security': 1, 'write_security': 1, 'preconditions': []}, {'did': 4105, 'name': 'DID-9', 'length_bytes': 4, 'encoding': 'bytes', 'read_sessions': [1, 3], 'write_sessions': [3], 'read_security': 1, 'write_security': 1, 'preconditions': []}]

dids[0].did: 4096

dids[0].name: "VIN"

dids[0].length_bytes: 4

dids[0].encoding: "bytes"

dids[0].read_sessions: [1, 3]

dids[0].write_sessions: [3]

dids[0].read_security: 1

dids[0].write_security: 1

dids[0].preconditions: [{"kind": "signal", "op": "==", "signal": "signal_0", "source_text": "signal_0 is true", "target": null, "unit": null, "value": true}]

dids[1].did: 4097

dids[1].name: "DID-1"

dids[1].length_bytes: 4

dids[1].encoding: "bytes"

dids[1].read_sessions: [1, 3]

dids[1].write_sessions: [3]

dids[1].read_security: 1

dids[1].write_security: 1

dids[1].preconditions: [{"kind": "signal", "op": "<", "signal": "vehicle_speed", "source_text": "vehicle_speed is below 10 km/h", "target": null, "unit": "km/h", "value": 10}]

dids[2].did: 4098

dids[2].name: "DID-2"

dids[2].length_bytes: 4

dids[2].encoding: "bytes"

dids[2].read_sessions: [1, 3]

dids[2].write_sessions: [3]

dids[2].read_security: 1

dids[2].write_security: 1

dids[2].preconditions: [{"kind": "signal", "op": "<", "signal": "battery_voltage", "source_text": "battery_voltage is below 14 V", "target": null, "unit": "V", "value": 14}]

dids[3].did: 4099

dids[3].name: "DID-3"

dids[3].length_bytes: 4

dids[3].encoding: "bytes"

dids[3].read_sessions: [1, 3]

dids[3].write_sessions: [3]

dids[3].read_security: 1

dids[3].write_security: 1

dids[3].preconditions: [{"kind": "signal", "op": "<", "signal": "response_delay", "source_text": "response_delay is below 50 ms", "target": null, "unit": "ms", "value": 50}]

dids[4].did: 4100

dids[4].name: "DID-4"

dids[4].length_bytes: 4

dids[4].encoding: "bytes"

dids[4].read_sessions: [1, 3]

dids[4].write_sessions: [3]

dids[4].read_security: 1

dids[4].write_security: 1

dids[4].preconditions: [{"kind": "signal", "op": "<", "signal": "engine_temp", "source_text": "engine_temp is below 95 \u00b0C", "target": null, "unit": "\u00b0C", "value": 95}]

dids[5].did: 4101

dids[5].name: "DID-5"

dids[5].length_bytes: 4

dids[5].encoding: "bytes"

dids[5].read_sessions: [1, 3]

dids[5].write_sessions: [3]

dids[5].read_security: 1

dids[5].write_security: 1

dids[5].preconditions: [{"kind": "signal", "op": "==", "signal": "signal_5", "source_text": "signal_5 is true", "target": null, "unit": null, "value": true}, {"kind": "signal", "op": "==", "signal": "ignition_on", "source_text": "ignition_on is true", "target": null, "unit": null, "value": true}]

dids[6].did: 4102

dids[6].name: "DID-6"

dids[6].length_bytes: 4

```
dids[6].encoding: "bytes"
dids[6].read_sessions: [1, 3]
dids[6].write_sessions: [3]
dids[6].read_security: 1
dids[6].write_security: 1
dids[6].preconditions: []
dids[7].did: 4103
dids[7].name: "DID-7"
dids[7].length_bytes: 4
dids[7].encoding: "bytes"
dids[7].read_sessions: [1, 3]
dids[7].write_sessions: [3]
dids[7].read_security: 1
dids[7].write_security: 1
dids[7].preconditions: []
dids[8].did: 4104
dids[8].name: "DID-8"
dids[8].length_bytes: 4
dids[8].encoding: "bytes"
dids[8].read_sessions: [1, 3]
dids[8].write_sessions: [3]
dids[8].read_security: 1
dids[8].write_security: 1
dids[8].preconditions: []
dids[9].did: 4105
dids[9].name: "DID-9"
dids[9].length_bytes: 4
dids[9].encoding: "bytes"
dids[9].read_sessions: [1, 3]
dids[9].write_sessions: [3]
dids[9].read_security: 1
dids[9].write_security: 1
dids[9].preconditions: []
routines: [{'rid': 8192, 'name': 'Routine-0', 'control_types': [1, 2, 3], 'sessions': [3], 'security': 1, 'parameter_lengths': {'1': 1, '2': 0, '3': 2}, 'preconditions': []}, {'rid': 8193, 'name': 'Routine-1', 'control_types': [1, 2, 3], 'sessions': [3], 'security': 1, 'parameter_lengths': {'1': 1, '2': 0, '3': 2}, 'preconditions': []}, {'rid': 8194, 'name': 'Routine-2', 'control_types': [1, 2, 3],
```

'sessions': [3], 'security': 1, 'parameter_lengths': {'1': 1, '2': 0, '3': 2}, 'preconditions': []]

routines[0].rid: 8192

routines[0].name: "Routine-0"

routines[0].control_types: [1, 2, 3]

routines[0].sessions: [3]

routines[0].security: 1

routines[0].parameter_lengths: {"1": 1, "2": 0, "3": 2}

routines[0].preconditions: []

routines[1].rid: 8193

routines[1].name: "Routine-1"

routines[1].control_types: [1, 2, 3]

routines[1].sessions: [3]

routines[1].security: 1

routines[1].parameter_lengths: {"1": 1, "2": 0, "3": 2}

routines[1].preconditions: []

routines[2].rid: 8194

routines[2].name: "Routine-2"

routines[2].control_types: [1, 2, 3]

routines[2].sessions: [3]

routines[2].security: 1

routines[2].parameter_lengths: {"1": 1, "2": 0, "3": 2}

routines[2].preconditions: []

timing: {'p2_ms': 50, 'p2_star_ms': 5000, 's3_ms': 5000}

nrc_priority: [19, 18, 17, 127, 126, 51, 53, 36, 49, 34]